AERO ELASTICITY

PROFESSIONAL ELECTIVES-III

VII Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6AE43	PEC	3	0	0	3	40	60	100

UNIT-I INTRODUCTION

Introduction to Aero elasticity COLLARS Triangle, Aerodynamics and interactions of Structural and Inertial forces Static and Dynamic Aero Elasticity Phenomena. Simple Two dimensional idealization of flow, String Theory, Fredholm Integral equations of Second Kind Exact Solutions for simple rectangular wings.

UNIT-II ANLAYTICAL METHODS

Formulations of Structural Dynamics Equation and Coupling effects for panels and plates, generalized coordinates, Lagrange's Equations of motion Hamilton's Principle Orthogonality conditions. Static Aero elastic Studies Divergences, control reversal, Aileron reversal speed, Aileron efficiency, lift distribution, Rigid and elastic wings.

UNIT-III EXPERIMENTAL ANALYSIS & EQUATIONS OF AERO ELASTIC

Non-dimensional Parameters, stiffness criteria, dynamic mass balancing, model experiments and dimensional similarity, flutter analysis. Formulation of Aero elastic Equations for a Typical Section, Quasi Steady Aerodynamic derivatives, modal equations Galerkins method of analysis

UNIT-IV FLUTTER

Stability of motion of Continua Torsion flexure flutter, Solution of flutter determinant, method ofdetermining the classical flutter speed, Flutter Prevention and control.

UNIT-V AERO ELASTICITY APPLICATIONS

Application of Aero Elasticity in Engineering Problems, Galloping of transmission lines, flow induces vibrations of tall slender structures and suspension Budges.

Text Books:

1. Dewey H. Hodges, G. Alvin Pierce (2011), Introduction to Structural Dynamics and Aero Elasticity, 2nd edition, Cambridge University Press, UK.
2. Fung Y. C. (2008), An introduction to the Theory of Aero Elasticity, Dover Publications, USA.
3. Jan R. Wright (2008), Introduction to Aircraft Aero Elasticity and Loads, John Wiley, USA

Reference Books:

1. Raymond L. Bisplinghoff, Holt Ashely (2002), Principles of Aeroelasticity, Drovers Publications, USA.
2. Adamu Yebi (2010), Vibration Analysis of Cracked Composite Aircraft Wing Modeled as Shell, VMD Verlag, New Delhi.
3. E. H. Dwell (1995), A Modern Course in Aero elasticity, Springer Publishers, Germany.

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. "Identify the aeroelastic phenomena flutter, divergence and aileron reversal arise and how they affect aircraft performance, collar triangle"
2. "Demonstrate a basic understanding of modern numerical methods and the state-of-the-art in structural dynamics and aeroelasticity"
3. Differentiate between static aeroelasticity and dynamic aeroelasticity
4. Analyze the wing flutter, under the over damping and critical damping conditions
5. Build confidence for self learning needed for aircrafts, automobiles failures due to vibration effect